



SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
CONTINUOUS ASSESSMENT TEST - I
FALL SEMESTER 2026-2027

SLOT: B2+TB2

Programme Name & Branch : M. Tech (MAI & MCS)
Course Code and Course Name : MACSE501 – Data Structures and Algorithm Analysis
Faculty Name : Parveen Sultana H, Rhymend Uthariaraj V
Class Number(s) : VL2026270106047, VL2026270106057
Date of Examination : 10/8/2026
Exam Duration : 90 minutes
Maximum Marks: 50

General instruction(s):

- Answer All Questions
- M - Max mark; CO – Course Outcome; BL – Blooms Taxonomy Level (1 – Remember, 2 – Understand, 3 – Apply, 4 – Analyse, 5 – Evaluate, 6 – Create)
- Course Outcomes:
CO1. Develop the ability to analyze the efficiency of algorithms using asymptotic notations and implement linear data structures.

CO2. Apply tree and graph-based data structures to design and implement efficient traversal, search, and pathfinding algorithms

Q. No	Questions	M	CO	BL
1.	Consider the following algorithm. Algorithm Remove Duplicates(A, n) <pre>count = 0 for i = 1 to n duplicate = false for j = 1 to i-1 if A[i] == A[j] duplicate = true end if end for if duplicate == false count = count + 1 end if end for return count</pre> a) Derive the computational time and auxiliary space complexity of the algorithm. Clearly explain each step of your analysis. b) Trace the algorithm for the input {8, 5, 8, 3, 5, 7, 9, 7}	5+5	1	BL3



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	and determine the final output. Explain how the execution supports the derived complexity.			
2.	An online shopping application maintains a list of customer orders(order name order ID) using a Singly Linked List. i) Explain the structure of a singly linked list with a neat labelled diagram. ii) Develop algorithms for the following operations. • Insert a new order at the end of the list. • Search for an order using its Order ID. • Delete an order with a given Order ID. Illustrate each operation with suitable diagrams. iii) State any two advantages of using a linked list over an array for managing customer orders.	2+6 +2	1	BL3
3.	Algorithm Compute(n) if $n \leq 1$ then return Compute(n-1) for $i \leftarrow 1$ to n do print(i) a) Formulate the recurrence relation for the running time. Explain the derivation. b) Solve the recurrence using the substitution method and determine the final time complexity in Big-O notation.	5+5	1	BL3
4.	a)Convert the following infix expression into its equivalent postfix expression using stack $((A+B)*C-(D/E+F)*(G-H))+I^J-K$ b) Using the postfix expression obtained in part (a), evaluate its value for the following data set using stack {A=11, B=10,C=9,D=8,E=7,F=6,G=5,H=4,I=3, J=2,K=1}	6+4	1	BL3



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5.	The following BST is given. <pre> 60 / \ 35 80 / \ / \ 20 50 70 90 / 45</pre> Without reconstructing the tree, answer the following. a) What is the search path for key 45? (2 Marks) b) Which node will replace 35 if it is deleted using the Inorder successor method? Draw the modified BST. (3 Marks) c) Where will the key 75 be inserted? Draw the updated BST. Obtain Inorder Traversal after the insertion.	2+3 +5	2	BL3
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